Preparation techniques for unconventional sand samples

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ABSTRACT: In physical modelling the preparation of sand samples can be essential in obtaining high-quality and reproducible results. This is especially true for processes for which the micro-structure of the sample affects the process, such as CPT-research and experiments on backward erosion piping, an internal erosion mechanism whereby pipes develop in a granular material in the opposite direction of flow. Several techniques exist for the preparation of homogeneous and fully saturated samples consisting of very uniform sand types. However, for closer resemblance of the process in field conditions, a need existed to prepare homogeneous samples with a fine fraction, a layered sample or a combination of clay and sand. This paper discusses techniques for preparation of homogeneous and layered natural sands containing fines and layered samples consisting of different types of sands or sand and clay.

1 INTRODUCTION

Soil model preparation is an important aspect of physical modelling in geotechnics. The preparation of homogeneous and fully saturated samples at different densities is the basis of high-quality and reproducible results.

Sample preparation therefore has received considerable attention in the past. At Deltares (former Delft Geotechnics / GeoDelft) sample preparation has been investigated since 1934 for different kinds of tests, like triaxial testing and larger scale model testing at 1- and n-g level. The need for larger soil models than triaxial test samples asked for reproducible, cost- and time-effective and multi-purpose preparation methods which lead to homogeneous sand samples. For many purposes also fully saturated samples were needed. Standard preparation techniques meant for triaxial testing were no longer efficient or applicable at larger scale.

An approach to reach very dense sand samples, with relative density larger than approximately 80%, is layered compaction (Mulilis et al., 1975; Mitchell et al., 1976). In this approach the sprinkling of sand is combined with continuous tamping.

Van der Poel and Schenkeveld (1998) describe a method for the preparation of very homogeneous sand models with different relative densities between 30% and 70%, using the dynamic fluidization-compaction method. This method involves the sprinkling of dry sand in a high column of de-aired water, thereby reducing air inclusions and creating a loose (near max. porosity) and homogeneous model. Subsequent compaction with minor vertical shock waves leads to a temporary fluidization of the whole sand body after which drainage of excess pore water pressures will take place, leading to homogeneous compaction. This technique was developed for the preparation of homogeneous multi-layer models intended for CPT-research and is used for many purposes.

Rietdijk et al. (2010) describe the drizzle method. This method is essentially similar to the dynamic fluidization-compaction technique, but uses a shower head for raining in the sand, which has the advantage that the sand does not need to be dried before sample preparation. However, this method has fewer possibilities compared with the original method (e.g. not suitable for multi-layer models).

The dynamic fluidization-compaction technique and layered compaction techniques have been used for many validation studies that required homogeneous samples. Uniform (sieved) sand samples are very suitable for this purpose. However, natural deposits are generally not homogeneous or uniform. The characteristics of the soil vary from micro-scale (laminae, bedding) to macro-scale. For closer resemblance to reality it can be required to test sand samples with fines or sand samples with consistent variation of characteristics.

This paper comprehensively describes a few examples of preparation of samples with natural sands and intended variation of soil properties.

2 PREPARATION TECHNIQUES

This paper describes the used preparation techniques for three different soil models. Two of the soil models were prepared to investigate backward erosion piping, a failure mechanism for water-retaining structures with granular material in the foundation covered by a cohesive material. A head drop across the structure may cause shallow open spaces ('pipes') to form in the granular material in the direction opposite of the water flow. Ongoing pipe formation may result in the failure of the structure (Van Beek et al., 2011).

In order to simulate the backward erosion piping mechanism in laboratory conditions, experiments were performed in a small-scale sand box (with inner dimensions of 0.5 x 0.3 x 0.1 m).

After numerous experiments with homogeneous samples (Van Beek et al., 2011, Van Beek et al., 2015), the influence of fines and layering is assessed. The used preparation techniques for these samples are described in section 2.1 and 2.2.

Another soil model was prepared in order to investigate the influence of multiple thin clay layers on cone resistance in the intermediate sand layers. In practice the cone resistance is a measure for the relative density of the sand layers, but standard relations are no longer applicable for thin sand layers embedded in softer soil layers. For different layer thicknesses (configurations) this effect has been investigated by performing CPT's in a rigid wall cylinder (inner diameter of 0.6 m). The preparation technique developed for this purpose is described in section 2.3.

2.1 *Heterogeneous sand samples for investigation of backward erosion piping*

The structure of natural sand deposits is dictated by its depositional history. Dikes are often founded on river deposits or tidal deposits. Both types of deposits are characterized by a variation of sand properties due to current ripples. Campbell (1967) describes the different units within a sedimentary body: micro-scale laminae, laminasets, beds and bedsets, which determine its properties at different scales.

Backward erosion piping is a micro-scale process as the pipe is only a few grains in depth and width at the tip. The process is therefore likely to be sensitive to micro-scale variations in the soil sample. A first attempt has therefore been made to simulate variations in sand characteristics along the pipe path by creating a sequence of layers in horizontal directions (already attempted earlier by Van Beek et al., 2008 and Dolphen, 2014; new experiments reported by Negrinelli, 2015) and by creating micro-scale variation resulting from natural sedimentation of particles. Both methods are described in this section.

Homogeneous samples in previous tests were prepared using the dynamic fluidization-compaction technique and layered compaction techniques (in order to create fully saturated samples). For preparation of these samples the box was placed in vertical position (for testing the box was placed in horizontal position).

2.1.1 *Research about preparation methods for micro-scale heterogeneity*

A preliminary experimental investigation was performed in order to identify a suitable method for the preparation of a sand sample that reproduces micro-scale heterogeneity. A cylindrical column was used. The objective was to prepare a sand sample for the small-scale piping test consisting of a single type of sand that can present local differences of different sand properties in the path of the pipe. As the sample in the piping sand box is generally built up with the box in vertical position, horizontal layering during preparation resulted in a variation of properties in the path of the horizontal pipe due to encountering the different layers.

The first trial was performed using a uniform natural sand (Enschede sand, d_{50}=0.380 mm, d_{60}/d_{10}=1.6). In order to reproduce micro-scale heterogeneity, the sample was composed by multiple superimposed horizontal sand layers (5 cm). Each layer was created by raining homogenously the sand of one layer in water. At the same time it was compacted by continuously tamping in normal direction with a compaction foot (its dimensions were chosen related to the soil grain size) in order to achieve a relative density of at least 85%. It was tried to create the heterogeneity at the interface between two layers by waiting 2 minutes before pouring the sand of the following layer. During this time interval, deposition of sand in water was expected to lead to segregation of sand particles. This should result in accumulation of fine sand grains at the interface of two sand layers causing local variation in sand properties like grain size and relative density, which can affect permeability and piping erosion resistance of the sand sample. At the beginning of each separate layer a small amount of sand was poured without tamping because in this way it was expected that disturbance of the fines that are deposited at the interface between the two layers during the waiting time interval could be avoided. After drainage of the sample local varieties were observed, but the 5 cm layers couldn't be distinguished (figure 1), probably due to the fact the sand is too uniform. During drainage of the sample it is possible to visualize zones with different water retention capacity (depending inter alia on content of fines and porosity). However, it was not clear whether the heterogeneity has been caused by change in density or grain size (or both).



Figure 1. Trial to reproduce micro-scale heterogeneity with Enschede sand.



Figure 2. Picture during preparation of a heterogeneous sand sample (micro-scale heterogeneity).

In order to obtain a more heterogeneous sand sample, a test was performed using a less uniform sand type. The used sand was 'Itterbeck mixture 2' (d_{50}=0.143 mm; d_{60}/d_{10}=3.2), which was created by mixing different fractions of sieved sand. The sample was created by pouring sand in layers of thickness equal to 3 cm. First, sand was poured into the water without tamping to create the first layer. By waiting (10 min) before pouring the sand of the following layer, it was tried to create heterogeneity at the interface between two layers. After pouring the first two layers and waiting again for 10 min, the next layer was poured into the column and after that the underlying layer was compacted by means of tamping while the upper part of top layer was mixed and hold in suspension, because the pounder is able to penetrate into the upper part (after pouring the soil is still very loosely packed). This procedure was repeated in order to get more layers (by waiting 10 min between every layer). The result of this method was considered as a good example of micro-scale heterogeneity. This was not only caused by the accumulation of fines between sand layers, but also inside the layers there was alternation between zones with fine sand particles and zones where coarser particles seem to be prevalent. Their thickness was in the order of millimeters. The last method was applied for preparation of samples in the piping box (figure 2). The piping process indeed proved to be sensitive to the micro-structure of the sand (Negrinelli, 2015).



Figure 3. Continous tamping during preparation of a macro-scale heterogeneity sample.

2.1.2 *Small-scale test preparation: Macro-scale heterogeneity*

In order to reproduce macro-scale heterogeneity, a sand body was formed by fine sand with an intermediate zone of coarse sand (heterogeneity was created

by difference in grain sizes). In this way the sand box consists of three layers: a coarse sand layer in the middle and two fine sand layers at the upstream and downstream sides of the sand box. First of all, the fine sand that forms the upstream layer was poured. Then the coarse sand intermediate zone was created and at the end the box was completely filled with fine sand. The preparation of each sand layer was performed pouring the sand slowly and uniformly into the box filled with water combined with continuous tamping (figure 3).

The main issue during the preparation of the sand sample was that fine and coarse sand can be mixed at the interface between different layers as a result of tamping near the interface. For this reason, at the beginning of the creation of a new layer, a certain amount of its sand was poured above the interface between the two layers before compaction was performed. A second problem occured when the first fine sand layer was completed. Fines were present in suspension in the water above the sand layer. For this reason, the water above the created layer was removed and was substituted with clean de-aired water. In this way it was possible to avoid the deposition of fines at the interface and the mixing of fines with the coarse sand.

This method has been applied in several piping experiments (example given in figure 4), of which the results are described in Negrinelli (2015).

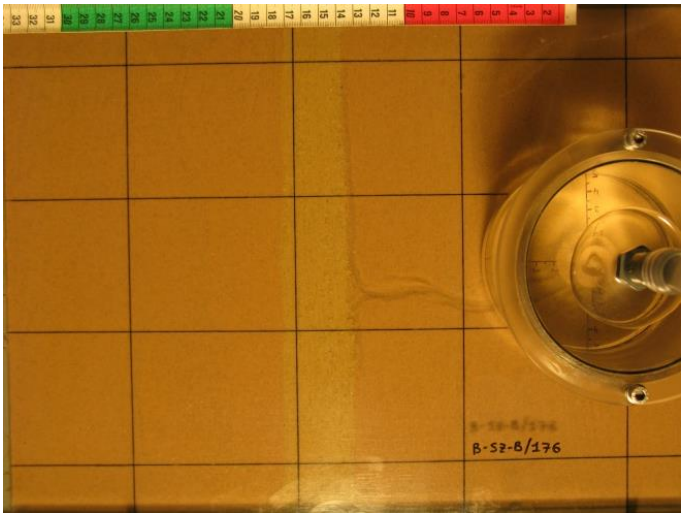


Figure 4. Example of pipe development in a heterogeneous sample (macro-scale heterogeneity).

2.2 *Homogeneous preparation of natural sands containing fines for investigation of backward erosion piping*

Uniform sand samples are fairly easy to prepare homogeneously, however difficulties may arise when a sample contains fines or is more graded. Homogeneous preparation of a sand containing fines using the methods in which the material is sprinkled in water, segregation of particles of different sizes and suspension of fines in the water above the layer may occur. Dry preparation methods can be applied, but full saturation can be difficult to achieve, especially when the sample consists of fine sand.

In the framework of FloodProBE (Benahmed & Philippe, 2012) backward erosion piping experiments needed to be performed with sand originating from a deposit near the Loire river (France) (d_{50}=0.365 mm, d_{60}/d_{10}=3.1, 6.6% with particle size smaller than 63 micron). Because the large amount of fines, it is hard to compact the material, because the fines will become a suspension with the water.

To overcome the issue of segregation, the material was rained into a very thin layer of water during simultaneous tamping. The thin layer of water was maintained by letting in de-aired water from below. A drawback of using a small water height is that air inclusions are more likely: the limited water height does not allow for separation of particles and loss of air as much as in a larger water height. After preparation, air inclusions are reduced by flushing de-aired water through the sample for 24 hours. Full saturation was not achieved in the experiments, which was visible during the experiment: some air was released as the pipe was formed.

2.3 *Model consisting of multiple (thin) clay and sand layers*

Soil models consisting of multiple thin Baskarp sand (d_{50}=0.151 mm, $_{d60/d10}$=1.5) and Vingerling clay (K147, undrained shear strength S_u= ±15 kPa, water content = 23.6%) layers have been prepared in order to investigate the influence of multiple thin soft layers on cone resistance in the intermediate layers (figure 6). Different configurations were tested in order to validate numerical simulations. The objective is to provide correction factors for the cone resistance in the intermediate sand layers, considering the sand layer thicknesses. This could result in a better estimate of the in situ relative density.

The dynamic fluidization-compaction and the layered compaction method were considered to be not applicable, since a sample height of more than 1 m and a relative density of around 50% were required (due to the rigid boundaries of the test cylinder). Using the dynamic fluidization-compaction method all relative densities up to 70% can be achieved, however, in order to achieve a homogeneous sample, the compaction must be carried out in one go. This seemed to be difficult when clay layers had to be included. Therefore an alternative layered compaction technique was used, which consisted of gently pouring the sand into the water up to a layer thickness of 5 cm had been reached and, after that, compacting that sand layer by gently tamping the sand layer with a flat cylindrical plate (diameter 6.5cm). The achieved relative density was controlled by measuring the height of the layer.

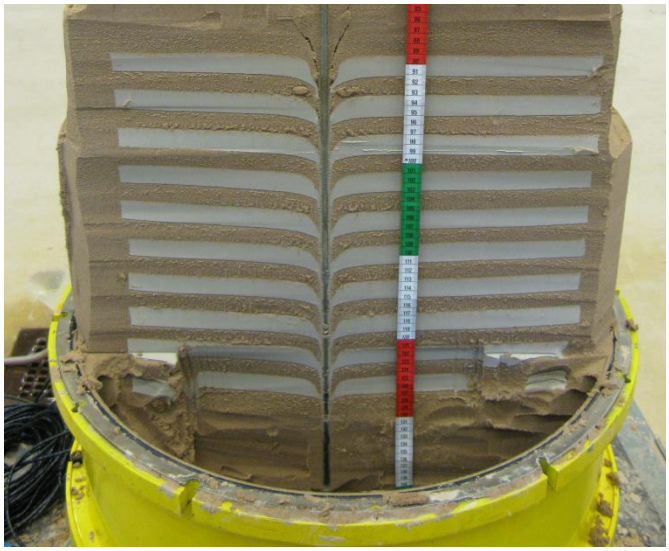
By lowering the water table below the level up to which the sand was prepared, capillary forces would

give extra strength to the sand surface in order to keep the sand intact when placing the layers of clay on top. Different configurations were tested: the layer thickness was respectively 2, 8 and 30 cm. The Vingerling clay was delivered in bricks of approximately 12.5x12.5x32 cm, which were trimmed into bricks of 10x10x25 cm with a steel wire, because of the fact that the bricks of clay were delivered sealed in plastic, in some cases damaged and/or partly dried out on the outside. With the same wire the thinner slices of clay were cut from the reshaped brick. The thin clay slices were lifted by using two suction cups on a plastic sheet sticking to the clay. After preparing the clay bricks, they were placed in the cell and pushed tightly together to prevent formation of air pockets between bricks as much as possible (figure 5).



Figure 5. Placement of the clay on top of the sand.

Figure 6. Cross section of soil model existing of multiple thin layers (2cm) after cone penetration testing.



Before using this technique on the actual samples, first a dummy layered sample was prepared with this technique. The bulk density of these layers seemed to be quite controllable and the positioning of the clay layers did not influence sand layer levels considerably and also no swell of the clay layers was observed after several days. The cone penetration tests in the prepared samples gave quite satisfying results and the achieved relative densities of different samples were comparable since the differences in measured cone resistances were within a range of 0.2 MPa (1 m penetration depth).

3 DISCUSSION

The examples illustrate that reproducible sample preparation is essential to physical modelling and that the mode of preparation will influence the final result.

The compaction method determines the range of relative density that can be achieved. Only samples with bulk relative densities of 0% and 100% are completely homogeneous. For each other value of the relative density it should be questioned whether the sample (bulk) density is the same as the local density. The preparation method must lead to the best fit between the bulk and the local density. When creating layered samples a constant relative density is most difficult to ensure, as continuous tamping may cause mixing of the different layers and dynamic fluidization-compaction must be carried out in one go.

The segregation of particles can be either unwanted or wanted and is controlled by the depth of the water height and applied compaction method.

The degree of saturation is also controlled by the preparation method: the success of saturation after dry preparation of the sample will depend on the soil type, its relative density and the content of fines, but full saturation will be difficult to achieve. When sprinkling sand in water, air inclusions can be prevented by drying the sand before sprinkling, using a sufficient water height and sprinkling at a low rate. De-aired water is preferred to tap water, due to the release of air from tap water in ambient conditions.

Preparation of layered samples or samples with fines can pose difficulties as a deviation is required from the preferred conditions. The examples in this paper show that each situation requires a different approach and that sample preparation requires experience, accuracy and competence.

4 CONCLUSIONS AND RECOMMENDATIONS

In literature several preparation techniques are described to create homogeneous samples using uniform sands. However, little information is available on preparation of non-homogeneous samples or less uniform samples (containing fines). To this end, several preparation techniques are discussed in this paper, including techniques for preparation of micro-scale heterogeneity, macro-scale heterogeneity in sand, preparation of sand samples containing fines and layering of clay and sand samples. The examples illustrate that sample preparation requires expe-

rience, accuracy and knowledge of the material behaviour.

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